# EXACTNESS OF THE CECH SEQUENCE OF A MODULE IS EQUIVALENT TO BIJECTIVITY OF THE COMPARISON MAP ON F-POWER TORSION

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 25364-25385 (canonical tag 0BNW) and the counted proof is the proof environment at source lines 25387-25423, which immediately follows it and is its own proof.

*Editorial note.* The statement refers to the displayed Cech sequence by its label equation-BL-cech-mod-re, which lies outside the counted spans; that label is carried as a navigation stub so that the reference resolves in the compiled document.

*Editorial note.* The two other references are to right exactness of the tensor product in another chapter and to the sibling lemma of this chapter; both are navigation stubs with their roles recorded in the reasons.

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*Allowed standard prerequisites (editorial).*

- The Cech sequence of a module for the pair consisting of a ring map and an element compares the module with its tensor product with the target ring and its localization. used at proof lines 25416-25419, where the sequence labelled equation-BL-cech-mod-re is referred to. Fixes the exact sequence whose exactness in each position is characterized by the lemma.
- Tensoring with a module is right exact, so the sequence in question is always exact on the right. invoked at proof lines 25389-25391. Gives part (1) immediately.
- The kernel of the localization map is the submodule of elements annihilated by a power of the element, which identifies the left exactness condition. used at proof lines 25393-25395. Gives part (2).

*Source reference.* Original equation 0F1R: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25346–25346. The reference points to the displayed Cech sequence that the statement and proof are about; it names the object under

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discussion and the element computations that decide exactness in each position are inside the counted span.

*Source reference.* Original lemma 00DF: sources/stacks-chow/tags.txt, source lines 495–495. The reference records right exactness of the tensor product, invoked at proof lines 25389-25391 for part (1) and again at proof lines 25416-25418 for the exactness in the middle for glueing pairs; it is a standard pointer and everything drawn from it is inside the counted span.

*Source reference.* Original lemma 0BNR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 25255–25255. The reference identifies the tensor product of the ring modulo its power torsion with the target modulo its own power torsion, used at proof lines 25388-25389 and 25399-25401; the exactness computation built on it is inside the counted span.

**Lemma 0.2.** *Let $R$ be a ring, let $f \in R$, and let $R \to R'$ be a ring map which induces isomorphisms $R/f^nR \to R'/f^nR'$ for $n > 0$. The sequence (0F1R) is*

1. *exact on the right,*
2. *exact on the left if and only if $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is injective, and*
3. *exact in the middle if and only if $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is surjective.*

*Thus $M$ is glueable for $(R \to R', f)$ if and only if $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is bijective. If $(R \to R', f)$ is a glueing pair, then $M$ is glueable for $(R \to R', f)$ if and only if $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is injective. For example, if $(R, f)$ is a glueing pair, then $M$ is glueable if and only if $M[f^\infty] \to (M \otimes_R R^\wedge)[f^\infty]$ is injective.*

*Proof.* We will use the results of Lemma 0BNR without further mention. The functor $M \otimes_R -$ is right exact (Algebra, Lemma 00DF) hence we get (1).

The kernel of $M \to M \otimes_R R_f = M_f$ is $M[f^\infty]$. Thus (2) follows.

If the sequence is exact in the middle, then elements of the form $(x, 0)$ with $x \in (M \otimes_R R')[f^\infty]$ are in the image of the first arrow. This implies that $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is surjective. Conversely, assume that $M[f^\infty] \to (M \otimes_R R')[f^\infty]$ is surjective. Let $(x, y)$ be an element in the middle which maps to zero on the right. Write $y = y'/f^n$ for some $y' \in M$. Then we see that $f^nx - y'$ is annihilated by some power of $f$ in $M \otimes_R R'$. By assumption we can write $f^nx - y' = z$ for some $z \in M[f^\infty]$. Then $y = y''/f^n$ where $y'' = y' + z$ is in the kernel of $M \to M/f^nM$. Hence we see that $y$ can be represented as $y'''/1$ for some $y''' \in M$. Then $x - y'''$ is in $(M \otimes_R R')[f^\infty]$. Thus $x - y''' = z' \in M[f^\infty]$. Then $(x, y'''/1) = (y''' + z', (y''' + z')/1)$ as desired.

If $(R \to R', f)$ is a glueing pair, then (0F1R) is exact in the middle for any $M$ by Algebra, Lemma 00DF. This gives the penultimate statement of the lemma. The final statement of the lemma follows from this and the fact that $(R, f)$ is a glueing pair if and only if $(R \to R^\wedge, f)$ is a glueing pair. □